A Kernel-Checked Conditional Reduction of Fefferman's Navier–Stokes Problem (A) to a Lipschitz Tether Rate and a Kato Maximal Solution

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Abstract

This paper records a **conditional** reduction of Fefferman's official Clay Millennium Problem (A) on $\mathbb{R}^3$ to two named hypotheses, together with a step-by-step derivation of every calculus identity in the reduction that the Lean 4 kernel has checked. The toolchain is Lean `v4.30.0-rc1` and Mathlib revision `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`. Every theorem printed in the reproducibility appendix depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`, and contains no `sorry`.

The main theorem, `clay_A_fefferman_of_rateLipschitz_of_kato`, states: if the Lipschitz / almost-everywhere form of the repaired tether rate, `FrohmanianTetherRateLipschitz`, holds for every Fefferman datum and every $\nu > 0$, and if `KatoMaximalSolution` holds for every such datum, then `Clay_A_Fefferman` holds. A parallel route, `clay_A_fefferman_of_rate_of_kato`, uses the earlier right-derivative form `FrohmanianTetherRateSmooth` in place of the Lipschitz form. As written, the two rate hypotheses are not comparable: a right derivative with only an upper bound need not be locally Lipschitz, and a Lipschitz function need not be right-differentiable at every time. A partial implication Smooth ⇒ Lipschitz is proved under an extra local bound on the right derivative.

Additional kernel-checked Kato infrastructure—including the Gagliardo–Nirenberg inequality and Kato–Ponce commutator estimate used by the H^3 energy bound, local H^s persistence on the H^3 existence interval, C^n regularity in time of the mild path, the identity $\partial_t u = \nu\Delta u - N(u)$ in $\widehat{H}^m$, joint C^2 regularity of the realized mild field (including the one-sided time derivative at $t = 0$), a Leray pressure, and the vanishing of $\int_{\mathbb{R}^3} \operatorname{div} F$ for a C^1 integrable field with integrable divergence—is recorded in Section 12. These results do not discharge `KatoMaximalSolution`.

This paper does not resolve any Clay Millennium Problem, and it does not claim unconditional three-dimensional Navier–Stokes regularity. The official statement of Problem (A) is reproduced unchanged. Both `FrohmanianTetherRateLipschitz` and `KatoMaximalSolution` remain unsettled for general Fefferman data (each is proved at the zero datum). The Beale–Kato–Majda logarithmic inequality is proved. The remaining analytic gaps in the rate are strain control along solutions and direction coercivity at a vorticity maximum.

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1 Introduction and non-claims

Fefferman's official statement of Clay Problem (A) [1] asks for a globally smooth, finite-energy solution of the incompressible Navier–Stokes equations on $\mathbb{R}^3 \times [0, \infty)$, with $f \equiv 0$ and $\nu > 0$, issuing from an arbitrary smooth, divergence-free initial datum that satisfies the pointwise decay (4). This paper does not prove that statement. It proves that the statement follows from two named hypotheses, and it records, with a Lean theorem name, file, line and commit next to each step, every identity in the argument that has been kernel-checked.

What is proved.

(1) A literal Lean transcription `Clay_A_Fefferman` of Fefferman's Problem (A) (Section 3).

(2) The Lipschitz / almost-everywhere rate `FrohmanianTetherRateLipschitz`, together with `KatoMaximalSolution`, implies `Clay_A_Fefferman` (Theorem 4.3; `clay_A_fefferman_of_rateLipschitz_of_kato`, `Scratch/TetherRateLipschitz.lean:251`, commit `5d8e6d6`).

(3) The earlier Smooth form of the rate, together with the same Kato hypothesis, also implies `Clay_A_Fefferman` (Theorem 4.4; `clay_A_fefferman_of_rate_of_kato`, `Scratch/LogBoundLocal.lean:719`).

(4) A comparison lemma that turns a locally Lipschitz cubic differential inequality holding almost everywhere into a uniform ceiling $M(t) \le \max\bigl(M(0), C/\kappa\bigr)$ (Section 5).

(5) The Beale–Kato–Majda logarithmic inequality on $\mathbb{R}^3$, as a theorem, not a hypothesis (Section 11; `bkmLogInequality`, `Scratch/LogBoundLocal.lean:659`).

(6) The evolution identities for $|\omega|$ at a spatial maximum, the algebraic assembly of the cubic rate, parabolic scaling of M, attainment of the vorticity maximum under decay at infinity, a shear counterexample to attainment without decay, and Lipschitz / Rademacher / Danskin regularity of M under a uniform bound on $\partial_t \omega$ (Sections 6–9).

(7) A kernel-checked refutation of a universal pointwise strain bound, and a kernel-checked refutation of the original (unrepaired) tether rate (Sections 10 and 15).

(8) A body of Kato-type infrastructure on the Fourier-side spaces H^s: the Leray projector and heat semigroup, equality of Fourier and spatial H^s energies, heat smoothing, Duhamel estimates, a tame product estimate for $s \ge 3$, the bilinear map NSN : $H^s \to H^{s-1}$, local mild existence and uniqueness in H^3, local persistence of H^s regularity on the H^3 existence interval, C^n regularity in time of the mild path in $\widehat{H}^m$, the identity $\partial_t u = \nu \Delta u - N(u)$ in those spaces, joint C^2 regularity of the realized mild field (including the one-sided time derivative at $t = 0$), a Gagliardo–Nirenberg inequality and Kato–Ponce commutator estimate, a Leray pressure, and the vanishing of $\int_{\mathbb{R}^3} \operatorname{div} F$ under an integrability hypothesis (Section 12).

What remains unsettled.

(1) `FrohmanianTetherRateLipschitz` (and likewise `FrohmanianTetherRateSmooth`) for general Fefferman data. Both are proved at $u_0 = 0$.

(2) `KatoMaximalSolution` for general Fefferman data. It is proved at $u_0 = 0$. Local persistence of H^s regularity on the H^3 existence interval and the Kato–Ponce commutator estimate are proved. The remaining formal pieces are listed in Section 12: joint C^∞ regularity of the realized mild field, the pointwise incompressible momentum equation with a pressure, the energy inequality in the Fefferman class, the energy right-derivative formula, the vanishing-flux package `H3IBPVanishingFlux`, and assembly of the maximal interval with the H^3 blow-up alternative.

(3) `StrainControlledByVorticity` (inequality 3) and `DirectionCoercivityAtMax` (inequality 4), along Navier–Stokes solutions.

(4) Propagation of Fefferman decay (or vanishing of vorticity at infinity) from $t = 0$ to later snapshots.

The gaps ledger T1–T10 from the earlier strain-kernel record [7] is restated in Section 14. None of those ten items is marked proved or disproved by the Lean work recorded here.

No sentence in this paper should be read as a claim that three-dimensional Navier–Stokes solutions are regular, or that Clay Problem (A) is solved.

2 Setting and conventions

Throughout, Sobolev spaces are written H^s, where $s \in \mathbb{N}$ is a nonnegative integer. (The letter m is reserved for miscellaneous integers and is never used as a Sobolev index in the prose.) In the Lean development the same spaces appear as `R3BKM.sobolevEnergy s` (spatial energy) and as the Fourier-side Hilbert space `Hm s`. When a theorem requires a definite integer threshold, that threshold is written explicitly (for instance $s = 3$, or $s \geq 3$).

In Lean, $\mathbb{R}^3$ is `T3 := EuclideanSpace ℝ (Fin 3)`. Velocity fields are maps `T3 → EuclideanSpace ℝ (Fin 3)`, and time-dependent fields are maps `ℝ → T3 → \(\ldots \)`. Spatial operators (`convective`, `laplacian`, `div`, `curl`, `pressureGradient`) are defined through Mathlib's Fréchet derivative `fderiv`, and ordinary time derivatives through `deriv`. Mathlib follows the standard convention that `fderiv ℝ f x = 0` and `deriv f x = 0` when f is not differentiable at x. A real supremum $\sup_x \|\omega(x)\|$ (Lean `iSup`) is 0 on an unbounded range. This is why several propositions carry an explicit `BddAbove` conjunct.

Write

$$M(t) \;:=\; \|\omega(t)\|_\infty \;:=\; \sup_{x \in \mathbb{R}^3} \big|\omega(x,t)\big|,$$

which is `RepairedTether.Mω` in `Scratch/RepairedTetherRate.lean:68`. The viscous coefficient is ν, and the forcing is identically zero.

Every mathematical claim in the body is either (i) a kernel-checked Lean theorem, cited by name, file, line and commit, or (ii) explicitly labelled as unformalized background (for example, a classical citation of Kato [3] or an analytic test that has no Lean theorem in the cited commits).

The Lean sources for the rate, the comparison lemma, attainment, and the Clay reduction live on the branch `feat/tether-rate-derivation` of `github.com/BenFrohman/frohmanian-tether-sdiff` at commit `5d8e6d679666d4cbd785326422695165655ab1ef` (the Lipschitz rate was introduced at `c9ffdeb1bfe8b7252bf93e6bd97f5455ed5afff7`; the tip `5d8e6d6` updates docstrings only). The Kato infrastructure cited in Section 12 lives on the diverged branch `feat/r3-bkm-continuation` at commit `b497281f935f256d2a05372d5955296794f1d46d`. The two branches share the merge-base `5dd8458aa85cd118962b8b7ef2f37480cf2cd0f7`. Line numbers for files present on both branches were checked at both tips; the one exception recorded here is `InverseSobolev.contDiff_fourierInv_of_all_energy`, which is at line 377 of `Scratch/InverseSobolev.lean` on the Kato branch (the pin used below) and at line 282 on the rate branch. Persistence of H^s regularity on the H^3 existence interval is a kernel-checked theorem on the Kato branch (`PersistenceHm.persist_on_H3_time`, `FeffermanMild.persist_fefferman`), not a draft.

3 Fefferman's Problem (A), stated literally

Fefferman's official problem description [1] poses Problem (A) as follows. Take $\nu > 0$ and $n = 3$. Let u° be a smooth divergence-free vector field on $\mathbb{R}^3$ satisfying

$$\big|\partial_x^\alpha u^\circ(x)\big| \leq C_{\alpha K}\,(1 + |x|)^{-K} \quad \text{on } \mathbb{R}^3, \text{ for any } \alpha, K. \tag{4}$$

Take $f \equiv 0$. Then there exist smooth functions $p(x,t)$ and $u_i(x,t)$ on $\mathbb{R}^3 \times [0,\infty)$ that satisfy (1) (the momentum equation), (2) ($\operatorname{div} u = 0$), (3) ($u(x,0) = u^\circ(x)$), (6) ($p, u \in C^\infty(\mathbb{R}^3 \times [0,\infty))$) and (7) ($\int_{\mathbb{R}^3} |u(x,t)|^2\,dx < C$ for all $t \geq 0$).

The Lean transcription, from `Scratch/RepairedTetherRate.lean:23--50` at `5d8e6d6`, is the following. It is the conclusion of every reduction theorem in this paper, and it is not altered.

```
def FeffermanDecay (u₀ : VelocityField) : Prop :=
  ∀ (k : ℕ) (α : Fin k → Fin 3) (K : ℝ), ∃ C : ℝ, ∀ x : T3,
    ‖iteratedFDeriv ℝ k u₀ x (fun j => EuclideanSpace.single (α j) (1 : ℝ))‖ ≤
      C * (1 + ‖x‖) ^ (-K)

def FeffermanGlobalSolution (ν : ℝ) (u₀ : VelocityField)
    (u : TimeDependentVelocity) (p : TimeDependentPressure) : Prop :=
  ContDiffOn ℝ ∞ (fun q : T3 × ℝ => u q.2 q.1) (Set.univ ×ˢ Set.Ici (0 : ℝ)) ∧
  ContDiffOn ℝ ∞ (fun q : T3 × ℝ => p q.2 q.1) (Set.univ ×ˢ Set.Ici (0 : ℝ)) ∧
  ClayStatements.navier_stokes_eq_Ici u p ν (fun _ _ => 0) ∧
  (∀ x : T3, u 0 x = u₀ x) ∧
  ∃ C : ℝ, ∀ t ≥ (0 : ℝ), ∫⁻ x, ‖u t x‖ₑ ^ 2 < ENNReal.ofReal C

def Clay_A_Fefferman : Prop :=
  ∀ (ν : ℝ) (u₀ : VelocityField),
    0 < ν → ContDiff ℝ ∞ u₀ → (∀ x, div u₀ x = 0) → FeffermanDecay u₀ →
      ∃ (u : TimeDependentVelocity) (p : TimeDependentPressure),
        FeffermanGlobalSolution ν u₀ u p
```

The one-sided time derivative in `navier_stokes_eq_Ici` is `derivWithin (fun s => u s x) (Set.Ici 0)`, so the equation at $t = 0$ does not depend on values for $t < 0$. The energy condition (7) is a strict inequality on the lower Lebesgue integral, so the Mathlib convention that a nonintegrable nonnegative function has integral 0 cannot discharge it.

3.1 Comparison with the repository's earlier `Clay_A`

The earlier statement `FrohmanianTether.ClayStatements.Clay_A` differs from Fefferman's text in one substantive respect: it uses a two-sided `deriv` at $t = 0$, which depends on an extension to $t < 0$ that the statement leaves unconstrained, and which takes the junk value 0 if that extension is not differentiable at 0. The decay condition in `Clay_A` uses the operator norm of the whole k-th derivative with $K \in \mathbb{N}$ and the form $C/(1+|x|)^K$; `FeffermanDecay` transcribes each multi-index on coordinate unit vectors, with $K \in \mathbb{R}$ and $C(1+|x|)^{-K}$. The energy in `Clay_A` is a non-strict Bochner integral together with an `Integrable` conjunct. Only the first of these differences is substantive. The official statement used in this paper is exclusively `Clay_A_Fefferman`.

Sanity check. The conclusion of Problem (A) is satisfiable at $u_0 = 0$ by the zero solution (`clay_A_fefferman_instance_zero`, `RepairedTetherRate.lean:144`), and `feffermanDecay_zero` (`RepairedTetherRate.lean:125`) shows that $u_0 = 0$ satisfies (4).

4 The two rate hypotheses and the main conditional theorems

The local solution class on which both rates are posed is `IsFeffermanSolutionOn` (`RepairedTetherRate.lean:56--65`): joint C^∞ regularity of (u, p) on $\mathbb{R}^3 \times [0, T)$; the incompressible Navier–Stokes equations with $f \equiv 0$ and a one-sided time derivative at $t = 0$; the initial condition $u(\cdot, 0) = u_0$; and the energy inequality $\int_{\mathbb{R}^3} |u(t)|^2 \le \int_{\mathbb{R}^3} |u_0|^2$, written as a comparison of lower Lebesgue integrals. The Lean docstring of this definition mentions that the energy *equality* holds for classical solutions; the formal proposition is the inequality just written, not an equality.

4.1 The Lipschitz / almost-everywhere rate

Definition 4.1 (Lipschitz tether rate)**.** The proposition `FrohmanianTetherRateLipschitz` (`Scratch/TetherRateLipschitz.lean:44--52`, commit `5d8e6d6`) states: there exist constants

$C, \kappa > 0$, depending only on u_0 and ν, such that along every local Fefferman-class solution issuing from u_0 on $[0, T)$,

(i) $M(t)$ is finite for every $t \in [0, T)$ (the range of $|\omega(t)|$ is bounded above);

(ii) M is locally Lipschitz on $[0, T)$: for every $b \in [0, T)$ there is a Lipschitz constant of M on the compact interval $[0, b]$;

(iii) for almost every $s \in (0, T)$, M is two-sided differentiable at s and

$$M'(s) \;\le\; C\,M(s)^2 - \kappa\,M(s)^3. \tag{1}$$

This proposition is unsettled for general Fefferman data. It is proved at the zero datum: `frohmanianTetherRateLipschitz_zero` (`TetherRateLipschitz.lean:263`).

4.2 The Smooth / right-derivative rate

Definition 4.2 (Smooth tether rate)**.** The proposition `FrohmanianTetherRateSmooth` (`Scratch/RepairedTetherRate.lean:74--81`) states: there exist $C, \kappa > 0$, depending only on u_0 and ν, such that along every local Fefferman-class solution issuing from u_0 on $[0, T)$, M is finite and continuous on $[0, T)$, and at every $s \in [0, T)$ there is a right derivative d of M at s satisfying $d \le CM(s)^2 - \kappa M(s)^3$.

This proposition is likewise unsettled for general Fefferman data, and is proved at the zero datum: `frohmanianTetherRateSmooth_zero` (`RepairedTetherRate.lean:170`). Its statement is not edited in this work.

4.3 The two forms are not comparable as written

A finite right derivative equipped only with an upper bound need not be locally Lipschitz: a lower bound on the right derivative is missing. Conversely, Lipschitz regularity yields a two-sided derivative only almost everywhere (Rademacher), not a right derivative at every time, and in particular not at $t = 0$. Therefore, as written, neither `FrohmanianTetherRateSmooth` nor `FrohmanianTetherRateLipschitz` implies the other.

What is proved is a partial implication. If, in addition to `FrohmanianTetherRateSmooth`, the right derivative of M is locally bounded on every compact $[0, b] \subset [0, T)$, then the Lipschitz / almost-everywhere rate holds, with the same constants C, κ. This is `frohmanianTetherRateLipschitz_of_smooth_of_locallyBddRightDeriv` (`TetherRateLipschitz.lean:361`). The local bound is not part of `FrohmanianTetherRateSmooth`. As a statement about Fefferman solutions with no extra bound, the implication Smooth $\Rightarrow$ Lipschitz is unsettled.

4.4 Main theorems

Theorem 4.3 (Conditional reduction, Lipschitz route)**.** *Assume*

(a) for every $\nu > 0$ and every Fefferman datum u_0 (smooth, divergence-free, satisfying (4)), the proposition `FrohmanianTetherRateLipschitz` u_0 ν holds;

(b) for every such ν and u_0, the proposition `KatoMaximalSolution` ν u_0 holds.

Then `Clay_A_Fefferman` holds.

This is `clay_A_fefferman_of_rateLipschitz_of_kato` (`Scratch/TetherRateLipschitz.lean:251--258`, commit `5d8e6d6`). The proof uses the already-proved logarithmic inequality `bkmLogInequality` (Section 11) and the assembly `r3BKMContinuation_of_kato_of_log` (`Scratch/R3BKMContinuation.lean:272`): the rate produces a uniform ceiling on M along every local solution (Section 5), and that ceiling is exactly the premise that turns `KatoMaximalSolution` into a global Fefferman solution.

An equivalent packaging, with the classical continuation hypothesis left named rather than reduced to Kato plus the log inequality, is `clay_A_fefferman_of_frohmanianTetherRateLipschitz` (`TetherRateLipschitz.lean:239`).

Theorem 4.4 (Conditional reduction, Smooth route)**.** *The same conclusion `Clay_A_Fefferman` follows if, in Theorem 4.3, hypothesis (a) is replaced by `FrohmanianTetherRateSmooth` for every Fefferman datum and every $\nu > 0$.*

This is `clay_A_fefferman_of_rate_of_kato` (`Scratch/LogBoundLocal.lean:719--725`). The older packaging with an unreduced continuation hypothesis is `clay_A_fefferman_of_frohmanianTetherRateSmooth` (`RepairedTetherRate.lean:115`).

4.5 Non-vacuity, and a global form that cannot feed Problem (A)

Both rate hypotheses, the Kato hypothesis, Fefferman decay, and the conclusion of Problem (A) are inhabited at $u_0 = 0$: `frohmanianTetherRateLipschitz_zero`, `frohmanianTetherRateSmooth_zero`, `katoMaximalSolution_zero` (`R3BKMContinuation.lean:254`), `feffermanDecay_zero`, `r3BKMContinuationAt_zero`, `clay_A_fefferman_instance_zero`. The zero solution is used, via the energy inequality in `IsFeffermanSolutionOn`, to show that every local Fefferman-class solution from $u_0 = 0$ vanishes (`eq_zero_of_isFeffermanSolutionOn_zero`, `RepairedTetherRate.lean:149`).

A global-solution form of the Smooth rate, `FrohmanianTetherRateSmoothGlobal` (`RepairedTetherRate.lean:189`), quantifies only over globally defined smooth physical solutions. It holds automatically for any datum that has no such solution (`frohmanianTetherRateSmoothGlobal_of_no_global_solution`, `RepairedTetherRate.lean:201`). That form therefore carries no information about existence, and is not used in either reduction.

5 The comparison ODE lemma

Write $\varphi(y) = Cy^2 - \kappa y^3$ for the cubic majorant, with $C, \kappa > 0$.

Lemma 5.1 (Antitone majorant)**.** *If $C/\kappa \le x \le y$, then $\varphi(y) \le \varphi(x)$. In particular, if $C/\kappa < Z$, then $\varphi(Z) < 0$, and $\varphi(y) \le \varphi(Z)$ whenever $y \ge Z$.*

This is `majorant_vector_field_antitoneOn` (`TetherRateLipschitz.lean:57`) together with the existing sign lemma `AnalyticPipeline.majorant_vector_field_neg_of_gt`. Indeed, $\varphi(y) = y^2(C - \kappa y)$, the factor $C - \kappa y$ is nonpositive and nonincreasing on $[C/\kappa, \infty)$, and y^2 is nondecreasing on $[0, \infty)$, so the product is nonincreasing there.

Lemma 5.2 (Finite-horizon Riccati comparison, Lipschitz / a.e. form)**.** *Let $M : \mathbb{R} \to \mathbb{R}$ satisfy $M \ge 0$ on $[0, T)$. Suppose M is Lipschitz on every compact interval $[0, b] \subset [0, T)$, and*

$$M'(s) \;\le\; C\,M(s)^2 - \kappa\,M(s)^3$$

for almost every $s \in (0, T)$, at points of two-sided differentiability. Then, for every $t \in [0, T)$,

$$M(t) \;\le\; \max\bigl(M(0),\, C/\kappa\bigr). \tag{2}$$

This is `comparison_ode_stability_lipschitz` (`Scratch/TetherRateLipschitz.lean:82--219`, commit `5d8e6d6`). The argument is as follows.

Proof. The claim is immediate at $t = 0$. Fix $t \in (0, T)$ and write $Y := \max\big(M(0), C/\kappa\big)$. It is enough to show $M(t) \le Y + \varepsilon$ for every $\varepsilon > 0$. Set $Z := Y + \varepsilon$, so $C/\kappa < Z$ and therefore $\varphi(Z) < 0$ by Lemma 5.1. Also $M(0) \le Z$.

Let K be a Lipschitz constant of M on $[0, t]$. Then M is continuous on the compact interval $I = [0, t]$, and (being Lipschitz) is absolutely continuous on $[0, t]$. Suppose, for a contradiction, that $Z < M(t)$. Let

$$S := \big\{ s \in I : M(s) \le Z \big\}.$$

Continuity of M makes S closed in I, hence compact, and $0 \in S$, so S is nonempty. Let $a := \max S$. Then $a < t$ (since $M(t) > Z$), and $M(a) \le Z$. In fact $M(a) = Z$: if $M(a) < Z$, continuity would provide a point of S strictly to the right of a. Thus a is a last-crossing time of the barrier Z.

Absolute continuity and the fundamental theorem of calculus give

$$M(t) - M(a) = \int_a^t M'(s)\, ds.$$

The almost-everywhere bound $M' \le \varphi(M)$ holds on $(0, T)$, hence almost everywhere on (a, t). On that interval, $M(s) > Z$ (no point of S lies strictly to the right of a), so Lemma 5.1 yields $\varphi(M(s)) \le \varphi(Z)$. Therefore

$$M(t) - M(a) \le \int_a^t \varphi\big(M(s)\big)\, ds \le \int_a^t \varphi(Z)\, ds = (t - a)\, \varphi(Z) < 0,$$

since $t - a > 0$ and $\varphi(Z) < 0$. Hence $M(t) < M(a) = Z$, contradicting $M(t) > Z$. Thus $M(t) \le Z$ for every $\varepsilon > 0$, and $M(t) \le Y$. □

Theorem 5.3 (Ceiling from the Lipschitz rate)**.** *If* `FrohmanianTetherRateLipschitz` $u_0\ \nu$ *holds, then there is a single ceiling*

$$Y \;=\; \max\big(\|\omega(u_0)\|_\infty,\, C/\kappa\big)$$

such that, along every local Fefferman-class solution from u_0, one has $M(t) \le Y$ and the range of $|\omega(t)|$ is bounded, for every t in the existence interval.

This is `ceiling_of_rateLipschitz` (`TetherRateLipschitz.lean:224`). The identification $M(0) = \|\omega(u_0)\|_\infty$ uses $u(0) = u_0$.

The Smooth-form comparison, used in Theorem 4.4, is the pre-existing right-derivative lemma `TetheredLyapunov.comparison_ode_stability_Icc_right` (`FrohmanianTether/TetheredLyapunov.lean:1520`): if M is continuous on $[0, t]$ and has at every $s \in [0, t)$ a right derivative $d \le \varphi(M(s))$, then the same ceiling (2) holds. Its proof is a barrier comparison with a strictly negative right slope at any contact point with $Y + \varepsilon$, rather than an integral identity. The corresponding ceiling for the Smooth rate is `ceiling_of_rate` (`RepairedTetherRate.lean:99`).

6 Evolution identities for the vorticity maximum

This section unfolds the cubic inequality (1) into identities that close and inequalities that do not. The Lean development is `Scratch/TetherRateDerivation.lean` at commit `5d8e6d6`.

6.1 The vorticity equation in the interior

For a two-sided solution, the vorticity formulation of Navier–Stokes is the existing identity `vorticity_transport_at` (`FrohmanianTether/NS_Equations.lean:2162`):

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega. \tag{3}$$

On a Fefferman solution the time derivative in the momentum equation is one-sided at $t = 0$. For times $t \in (0, T)$, the one-sided derivative coincides with the two-sided derivative, because $[0, \infty)$ is a neighbourhood of t. This is `fefferman_two_sided_eq_interior` (`TetherRateDerivation.lean:431`), together with `derivWithin_Ici_eq_deriv_of_pos` and the interior smoothness lemmas `fefferman_u_contDiffAt_interior`, `fefferman_p_contDiffAt_interior` (`TetherRateDerivation.lean:411--425`). Status: kernel-checked proved. At $t = 0$ the transport identity must be read with a right derivative; that one-sided commutation of curl and ∂_t is not formalized in this file, and is not claimed here.

Take the inner product of (3) with ω and use

$$\partial_t\Big(\tfrac{1}{2}|\omega|^2\Big) = \langle \omega, \partial_t \omega \rangle,$$

which is `hasDerivAt_halfNormSq_time` (`TetherRateDerivation.lean:281`). One obtains the pointwise energy law

$$\partial_t\Big(\tfrac{1}{2}|\omega|^2\Big) + \langle \omega, (u \cdot \nabla)\omega \rangle = \langle \omega, (\omega \cdot \nabla)u \rangle + \nu \langle \omega, \Delta \omega \rangle. \tag{4}$$

6.2 The Bochner identity

Write $|\nabla \omega|^2_{\mathrm{HS}} = \sum_{i=1}^{3} |\partial_i \omega|^2$ (`TetherRateDerivation.hsSq`) and write Δ for the scalar Laplacian (`scalarLaplacian`). For a C^2 field,

$$\langle \omega, \Delta \omega \rangle = \Delta\big(\tfrac{1}{2}|\omega|^2\big) - |\nabla \omega|^2_{\mathrm{HS}}. \tag{5}$$

This is `bochner_identity` (`TetherRateDerivation.lean:78`). Status: kernel-checked proved.

6.3 At a spatial maximum

Suppose x is a local spatial maximum of $y \mapsto |\omega(y)|$. Then it is a local maximum of $\frac{1}{2}|\omega|^2$ (`isLocalMax_halfNormSq_of_isLocalMax_norm`, `TetherRateDerivation.lean:114`), the Fréchet derivative of $\frac{1}{2}|\omega|^2$ vanishes (`fderiv_halfNormSq_eq_zero_of_isLocalMax`, :127), and therefore the transport pairing vanishes:

$$\langle \omega, (u \cdot \nabla)\omega \rangle(x) = 0. \tag{6}$$

This is `transport_vanishes_at_local_max` (`TetherRateDerivation.lean:147`). Moreover, the scalar Laplacian of a C^2 function is nonpositive at a local maximum (`scalarLaplacian_nonpos_of_isLocalMax`, `TetherRateDerivation.lean:202`, from the one-dimensional second-derivative test `deriv_deriv_nonpos_of_isLocalMax`). Combined with (5),

$$\langle \omega, \Delta \omega \rangle(x) \;\le\; -\,|\nabla \omega(x)|^2_{\mathrm{HS}} \;\le\; 0. \tag{7}$$

This is `viscPair_le_neg_hsSq_at_local_max` (`TetherRateDerivation.lean:256`). Status of (6)–(7): kernel-checked proved.

6.4 Envelope identities for M

If $M(s) = |\omega(s, x(s))|$ along a differentiable maximizing trajectory, and if the spatial gradient of $\frac{1}{2}|\omega|^2$ vanishes at $x(t)$, the chain rule and (6) give, whenever $M(t) > 0$,

$$M'(t) = \frac{\langle \omega, \partial_t \omega \rangle}{M(t)}\,(t, x(t)), \tag{8}$$

provided the path derivative of $\omega(s, x(s))$ splits as $\partial_t \omega + (x' \cdot \nabla)\omega$. This is `envelope_along_differentiable_maximizer` (`TetherRateDerivation.lean:317`), using `hasDerivAt_norm_of_ne_zero` (:291). Status: kernel-checked proved, conditionally on those path hypotheses. Existence of a differentiable maximizing path is the named proposition `DifferentiableMaximizer` (`TetherRateDerivation.lean:454`), and is unsettled. The Lipschitz route in Sections 7–8 does not use such a path.

A difference-quotient form of Danskin's inequality, using a maximizer at time $t+h$ rather than at time t, is `danskin_difference_quotient` (`TetherRateDerivation.lean:351`). Status: kernel-checked proved. The comparison that uses only a maximizer at time t goes the wrong way and does not bound M' from above.

Combining (4), (6), (7) and an envelope formula at a spatial maximum with $M > 0$ yields

$$M' \;\le\; \frac{\langle \omega, (\omega \cdot \nabla)u \rangle + \nu \langle \omega, \Delta \omega \rangle}{M} \;\le\; \frac{\langle \omega, (\omega \cdot \nabla)u \rangle}{M} - \nu \frac{|\nabla \omega|^2_{\mathrm{HS}}}{M}. \tag{9}$$

6.5 The stretching term, aiming at CM^2

By Cauchy–Schwarz and the definition of the operator-norm strain proxy `strain_sup_norm`,

$$\langle \omega, (\omega \cdot \nabla)u \rangle \;\le\; \|\nabla u\|_\infty\, |\omega|^2 \;\le\; \|\nabla u\|_\infty\, M^2,$$

at any point, once the strain is bounded above. This is the existing identity `inner_convective_le_strain`, wrapped as `stretching_le_strain_mul_norm_sq` (`TetherRateDerivation.lean:363`). Status: kernel-checked proved, as linear algebra, not as a Calderón–Zygmund estimate.

To pass from this to a source CM^3, and hence after division by M to the quadratic term CM^2, one needs

$$\|\nabla u(t)\|_\infty \;\le\; C\, M(t) \tag{10}$$

uniformly in $t \in [0, T)$, with C depending only on u_0 and ν. This is the named proposition `StrainControlledByVorticity` (`TetherRateDerivation.lean:465`). Status: unsettled along Fefferman-class solutions. A universal bound of the same shape, imposed on arbitrary fields, is disproved in Section 10.

6.6 The dissipative term, aiming at $-\kappa M^3$

From (7), the viscous contribution to M' is at most $-\nu|\nabla\omega|^2_{\mathrm{HS}}/M$. At a spatial maximum of $|\omega|$, the polar decomposition $\nabla\omega = (\nabla|\omega|) \otimes \xi + |\omega|\nabla\xi$ and $\nabla|\omega| = 0$ give $|\nabla\omega|^2_{\mathrm{HS}} = M^2|\nabla\xi|^2_{\mathrm{HS}}$. The algebraic Pythagoras identities that record this reduction are `polar_pythagoras_directional_deriv` and `concrete_bochner_polar_viscous_damping_at_max` in `FrohmanianTether/SymplecticConstraint.lean:567` and :614. The cubic sink in (1) is then equivalent, after dividing by M and absorbing $\nu > 0$ into the constant, to a lower bound

$$|\nabla\omega|^2_{\mathrm{HS}} \;\ge\; \kappa\, M^4 \tag{11}$$

at the attained maximum, with $\kappa > 0$ independent of t. This is the named proposition `DirectionCoercivityAtMax` (`TetherRateDerivation.lean:474`). Status along Navier–Stokes trajectories: unsettled. Section 10 records why the same lower bound cannot be a universal static fact; that discussion is labelled as unformalized background where no Lean theorem exists.

6.7 Algebraic assembly

Lemma 6.1 (Cubic assembly)**.** *Suppose $M > 0$, $\nu \geq 0$, $M' \leq \big(\text{stretch} + \nu \cdot \text{visc}\big)/M$, $\text{stretch} \leq CM^3$, $\text{visc} \leq -|\nabla\omega|^2_{\text{HS}}$, and $\kappa M^4 \leq |\nabla\omega|^2_{\text{HS}}$. Then*

$$M' \;\leq\; CM^2 - \nu\kappa\, M^3.$$

This is `cubic_rate_of_strain_and_hsSq` (`TetherRateDerivation.lean:390`), built from `cubic_rate_of_stretching_and_visc` (:373). Status: kernel-checked proved, as an implication. It does not discharge the two inequalities it assumes. For $\nu > 0$ the factor $\nu\kappa$ may be renamed as the constant in (1); for $\nu = 0$ the cubic sink produced by this assembly vanishes. Clay Problem (A) has $\nu > 0$. The wrapper `cubic_rate_from_named_hypotheses` (:609) records the same implication and does not claim the hypotheses or prove either rate proposition for general data.

7 Attainment of the vorticity maximum, and a shear counterexample

Write `VorticityMaximumAttained` u T $x_{\max}$ for the statement that for every $t \in [0, T)$, the point $x_{\max}(t)$ is a global spatial maximizer of $y \mapsto |\omega(y, t)|$ (`TetherRateDerivation.lean:448`). As a claim about every solution of class `IsFeffermanSolutionOn`, this is false. Under a decay hypothesis, it is true.

Lemma 7.1 (Attainment for a continuous function vanishing at infinity)**.** *A continuous nonnegative function $f : \mathbb{R}^3 \to \mathbb{R}$ that tends to 0 at infinity attains a global maximum.*

This is `exists_isMaxOn_of_tendsto_zero` (`Scratch/VorticityMaxAttained.lean:58`, commit `5d8e6d6`). Nonnegativity is used: a negative function approaching 0 need not attain its supremum 0. If f vanishes identically the claim is trivial. Otherwise pick x_0 with $f(x_0) > 0$; vanishing at infinity supplies a compact set outside which $f < f(x_0)$; adjoining x_0 yields a nonempty compact on which a continuous function attains its maximum, and that maximum is global.

Lemma 7.2 (Decay of vorticity under Fefferman decay)**.** *If u is C^1 and satisfies `FeffermanDecay`, then $|\omega(x)| \to 0$ as $|x| \to \infty$.*

This is `vorticity_tendsto_zero_of_feffermanDecay` (`VorticityMaxAttained.lean:269`). The proof bounds $|\text{curl}\, u|$ by six times the operator norm of ∇u (`norm_curl_le_six_fderiv`, :164), applies `FeffermanDecay` at order one to obtain $|\nabla u(x)| \leq C(1 + |x|)^{-1}$, and squeezes to zero at infinity. Consequently, a C^1 snapshot with Fefferman decay attains a global maximum of $|\omega|$ (`exists_isMaxOn_vorticity_of_feffermanDecay`, :297). An attained maximum is a genuine L^∞ representative: the range is bounded and $\|\omega\|_\infty = |\omega(x)|$ (`isMaxOn_vorticity_sup_norm_eq`, :95).

Theorem 7.3 (Attainment under snapshot decay)**.** *Let (u, p) be a local Fefferman-class solution on $[0, T)$. If every snapshot $u(t)$ has continuous $|\omega(t)|$ vanishing at infinity, then there exists a (not necessarily measurable) map $x_{\max}$ such that `VorticityMaximumAttained` u T $x_{\max}$ holds. The same conclusion holds if every snapshot satisfies `FeffermanDecay`.*

These are `exists_vorticityMaximumAttained_of_tendsto_zero` (`VorticityMaxAttained.lean:321`) and `exists_vorticityMaximumAttained_of_snapshot_feffermanDecay` (:335). Joint Fefferman smoothness implies that every snapshot is spatially C^∞ (`fefferman_snapshot_contDiff`, :307).

Theorem 7.4 (Attainment at time zero)**.** *If (u, p) is a local Fefferman-class solution on an interval with $0 < T$, and if the initial datum u_0 satisfies `FeffermanDecay`, then $|\omega(0)|$ attains a global maximum.*

This is `exists_isMaxOn_vorticity_at_zero` (`VorticityMaxAttained.lean:350`). Status: kernel-checked proved. Propagation of Fefferman decay, or of vanishing at infinity, from $t = 0$ to $t > 0$ is not proved here. That propagation is one of the remaining pieces of the Kato track (Section 12): once a solution is known to remain in every H^s, vanishing at infinity is the expected consequence of the embedding and the decay of the initial datum, but it is not a theorem in the cited commits.

Theorem 7.5 (The solution class does not imply attainment)**.** *It is not the case that every* `IsFeffermanSolutionOn` *solution admits a maximizing trajectory. An explicit counterexample is the stationary shear*

$$u(x) \;=\; \big(y \arctan y - \tfrac{1}{2}\log(1+y^2),\, 0,\, 0\big), \qquad p \equiv 0, \qquad \nu = 0,$$

where $y = x_2$. Its vorticity is $\omega = (0, 0, -\arctan y)$, which never attains the value $\pi/2 = \sup|\omega|$.

This is `not_isFeffermanSolutionOn_implies_vorticityMaximumAttained` (`VorticityMaxAttained.lean:569`), using `isFeffermanSolutionOn_arctanShear` (:545) and `not_isMaxOn_abs_arctan` (:484). The field is C^∞ and divergence-free, the convective term and the pressure gradient vanish, and the energy inequality holds as an equality of two (possibly infinite) lower integrals. The class `IsFeffermanSolutionOn`, as written, does not require $\nu > 0$, spatial decay, or finite energy. The counterexample therefore does not touch the Clay reduction, which assumes $\nu > 0$ and Fefferman data. It shows only that attainment cannot be read off from the local solution predicate alone.

8 Lipschitz regularity, Rademacher, and Danskin's formula for M

A maximizer itself need not move smoothly: two peaks can trade places, and a selected maximum point can jump. The Lipschitz rate does not need a smooth maximizer. What is used is regularity of the scalar envelope M.

The hypothesis that makes M Lipschitz is a uniform-in-space bound on the time derivative of vorticity. Write `VorticityTimeDerivBddOn` $u\ I\ C$ for the statement that $C \geq 0$ and, for every $t \in I$ and every $x \in \mathbb{R}^3$, the map $s \mapsto \omega(x, s)$ is differentiable at t with $|\partial_t \omega(x, t)| \leq C$ (`VorticityMaxAttained.lean:586`). This bound is *not* implied by `IsFeffermanSolutionOn`: joint C^∞ regularity on $\mathbb{R}^3 \times [0, T)$ does not by itself give bounds uniform in x.

Lemma 8.1 (Lipschitz envelope)**.** *If* `VorticityTimeDerivBddOn` $u\ [a, b]\ C$ *holds and $|\omega(t)|$ is bounded for each $t \in [a, b]$, then*

$$\big|M(t) - M(s)\big| \;\leq\; C\,|t - s| \qquad \text{for all } s, t \in [a, b].$$

In particular, M is Lipschitz on $[a, b]$ with constant C.

These are `abs_Mω_sub_le_of_timeDerivBdd` (`VorticityMaxAttained.lean:592`) and `lipschitzOnWith_Mω_of_timeDerivBdd` (:631). The pointwise estimate $|\omega(x, t) - \omega(x, s)| \leq C|t - s|$ is the mean value inequality at each x; taking suprema, using boundedness of each snapshot so that M is a genuine supremum, yields the claim for M.

Lemma 8.2 (Rademacher)**.** *Under the same hypotheses, with $a < b$, the map M is differentiable at almost every $t \in (a, b)$. At those interior points the derivative is two-sided.*

This is `ae_differentiableAt_Mω_of_timeDerivBdd` (`VorticityMaxAttained.lean:644`), applying Mathlib's Rademacher theorem on the compact interval.

Theorem 8.3 (Danskin equality at differentiability points)**.** *Suppose M is two-sided differentiable at a time t, a spatial maximum of $|\omega(t)|$ is attained at a point x, $M(t) > 0$, the map $s \mapsto \omega(x, s)$ is differentiable at t, and $|\omega(x, s)| \le M(s)$ for s near t. Then*

$$M'(t) = \frac{\langle \omega(x,t),\, \partial_t \omega(x,t) \rangle}{M(t)}. \tag{12}$$

In particular, the inequality `DanskinRateAt` *holds at t.*

This is `danskin_eq_of_differentiable` (`VorticityMaxAttained.lean:671`). The proof is: the map $s \mapsto |\omega(x,s)|$ has derivative equal to the right-hand side of (12) by `hasDerivAt_norm_of_ne_zero`; the difference $M(s) - |\omega(x,s)|$ is nonnegative near t and zero at t, hence has a local minimum at t, so its derivative vanishes; therefore $M'(t)$ equals the derivative of $|\omega(x,s)|$.

Under `VorticityMaximumAttained`, the near-time comparison $|\omega(x,s)| \le M(s)$ holds automatically at interior times (`eventually_norm_le_Mω_of_attained`, :721). Consequently `DanskinRateAt` holds at every interior differentiability point of M at which $s \mapsto \omega(x_{\max}(t), s)$ is differentiable (`danskinRateAt_of_differentiable`, :734), and therefore almost everywhere on compact subintervals of $(0,T)$ under a uniform bound on $\partial_t \omega$ (`ae_danskinRateAt_of_timeDerivBdd`, :751).

Two narrowings of these statements are recorded here rather than papered over. First, `ae_danskinRateAt_of_timeDerivBdd` takes as a hypothesis the time-differentiability of $s \mapsto \omega(x_{\max}(t), s)$ at each t in the interval. For each fixed t the point $x_{\max}(t)$ is frozen, so on a Fefferman solution in the open interval $(0,T)$ this is differentiability in time of a spatially smooth field at a single point, which interior joint smoothness supplies; that discharge is not part of the Lean theorem as stated. Second, the uniform bound `VorticityTimeDerivBddOn` is not proved for Fefferman-class solutions in the cited commits. The Lipschitz regularity of M along a general local solution remains a part of the open rate hypothesis, not a theorem about every `IsFeffermanSolutionOn` solution.

9 Parabolic scaling of M

The Navier–Stokes scaling $u_\lambda(x,t) = \lambda\, u(\lambda x, \lambda^2 t)$ is `rescaleFlow` (`TetherRateDerivation.lean:493`). Under componentwise spatial differentiability,

$$\omega_\lambda(x,t) = \lambda^2\, \omega(\lambda x, \lambda^2 t),$$

which is `vorticity_rescaleFlow` (:501). Taking suprema, for $\lambda \neq 0$ and a bounded snapshot,

$$M(u_\lambda, t) = |\lambda|^2\, M(u, \lambda^2 t). \tag{13}$$

This is `Mω_rescaleFlow` (`TetherRateDerivation.lean:513`). Differentiating the envelope $s \mapsto \lambda^2 M(\lambda^2 s)$ at a point where M has derivative d at $\lambda^2 t$ gives, for $\lambda > 0$,

$$M'_\lambda(t) = \lambda^4\, d = \lambda^4\, M'(\lambda^2 t). \tag{14}$$

This is `hasDerivAt_Mω_rescale_of_hasDerivAt` (`TetherRateDerivation.lean:548`). Status: kernel-checked proved.

The cubic rate is not scaling-invariant with a fixed pair (C, κ). The identity `rate_form_rescale_iff` (`TetherRateDerivation.lean:566`) states that, for $\lambda > 0$,

$$\lambda^4 d \le C\big(\lambda^2 M\big)^2 - \kappa\big(\lambda^2 M\big)^3 \qquad \text{if and only if} \qquad d \le CM^2 - (\kappa\lambda^2) M^3,$$

where M and d are evaluated at $\lambda^2 t$. The quadratic source term scales the same way as M'_λ. The cubic sink is one degree supercritical: a pair (C, κ) that works for u_λ corresponds to a sink coefficient $\kappa\lambda^2$ for u. For concentrated data (large λ), any κ that is to work uniformly must shrink at least like λ^{-2}. This is a constraint on the open coercivity inequality, not a proof of it.

10 The universal strain bound is disproved; static coercivity is not a general fact

Theorem 10.1 (No universal pointwise strain bound)**.** *For every constant $C \in \mathbb{R}$, it is not the case that*

$$\|\nabla u(x)\| \;\le\; C\,\|\operatorname{curl} u\|_\infty$$

holds for every velocity field u and every $x \in \mathbb{R}^3$.

This is `StrainBoundRefutation.not_forall_strain_le_sup_vorticity` (`FrohmanianTether/StrainBoundRefutation.lean:78`, commit `5d8e6d6`). The counterexample is the linear pure-strain (irrotational) field $u(y) = (y_1, y_0, 0)$: $\operatorname{curl} u = 0$, so the right-hand side is zero, while ∇u is a nonzero constant. The field does not decay and is not recovered from vorticity by the Biot–Savart law, so the refutation does not by itself disprove `StrainControlledByVorticity` on Fefferman snapshots. On decaying divergence-free fields the map $\omega \mapsto \nabla u$ is a Calderón–Zygmund operator, which is unbounded on L^∞; the standard substitute is a logarithmic estimate involving a higher norm (Section 11). Whether a constant $C = C(u_0, \nu)$ nevertheless controls $\|\nabla u\|_\infty / M$ uniformly in time along every local Fefferman solution is unsettled.

Rigid rotation, labelled unformalized background. The same lower bound (11), imposed on arbitrary smooth snapshots rather than on a Navier–Stokes trajectory, is not a general fact about maxima of $|\omega|$. A rigid rotation $u(x) = (-x_2, x_1, 0)$ has constant vorticity $\omega = (0, 0, 2)$, so every point is a maximum, $M = 2$, and $|\nabla\omega|_{\mathrm{HS}} = 0$. The dissipation term vanishes identically, and (11) fails for every $\kappa > 0$. This field does not decay and is not a Fefferman datum; the example shows only that (11) cannot be a universal static inequality, so if it holds along a solution it must use the dynamics. A wide, nearly rigid Gaussian vortex $u(x) = e^{-\varepsilon|x|^2}(-x_2, x_1, 0)$ produces the same vanishing of $|\nabla\omega|$ at the origin and a dissipation of size $O(\varepsilon)$, with $M = 2$ fixed as $\varepsilon \to 0$.

These two calculations are recorded here as an analytic test. They are **not** kernel-checked Lean theorems in the cited commits. The file `Scratch/TetherRateStaticTests.lean`, named in the docstring of `DirectionCoercivityAtMax` as containing a kernel-checked static disproof, is not present in either cited worktree. The status of static coercivity, as a Lean statement, is therefore: not formalized, not proved, and not disproved, in the commits used here. The status of `DirectionCoercivityAtMax` along Navier–Stokes trajectories remains unsettled.

11 The Beale–Kato–Majda logarithmic inequality

Theorem 11.1 (BKM logarithmic inequality)**.** *There exists a constant $C > 0$ such that for every smooth divergence-free field $f : \mathbb{R}^3 \to \mathbb{R}^3$ with finite H^3 energy, the gradient ∇f is bounded, and*

$$\|\nabla f\|_\infty \;\le\; C\Big(1 + \|f\|_2^2 + \|\operatorname{curl} f\|_\infty\big(1 + \log\big(1 + \|f\|_{H^3}^2\big)\big)\Big). \tag{15}$$

This is `LogBoundLocal.bkmLogInequality` (`Scratch/LogBoundLocal.lean:659`, commit `5d8e6d6`), proving the proposition `R3BKM.BKMLogInequality` as stated at `R3BKMContinuation.lean:63`. The Lean proposition asserts boundedness of ∇f and the inequality (15); boundedness of $\operatorname{curl} f$ is not a separate conjunct of the proposition, and is obtained in the proof from the bound on ∇f (`GradBound.bddAbove_norm_fderiv`, `Scratch/GradBound.lean:112`, together with $|\operatorname{curl} f| \le 6\|\nabla f\|$). The integer $s = 3$ is the index used in the formal statement (classically, any $s > 5/2$). Status: kernel-checked proved.

The argument, as assembled in `Scratch/LogBound.lean` and `Scratch/LogBoundLocal.lean`, is as follows.

Proof outline, matching the Lean development. Boundedness of the gradient from H^3 energy. `GradBound.bddAbove_norm_fderiv` produces a bound on $|\nabla f(x)|$ from the finite sum of L^2 norms of derivatives that constitutes `sobolevEnergy 3 f`. Thus ∇f is bounded, and so is $\operatorname{curl} f$.

Localization. Fix a point x_0 and a scale $L > 0$. A standard bump φ_L equals 1 on the ball of radius L about x_0, is supported in a fixed multiple of that ball, and has derivative bounds scaling as L^{-k} (`LogBoundLocal.φL` and the lemmas following `LogBoundLocal.lean:205`). The derivatives of f at x_0 agree with those of the compactly supported field $\varphi_L f$.

Symbol decomposition. Each second-order Riesz-type symbol $m_{kl}(\xi) = \xi_k\xi_l/|\xi|^2$ is split, after multiplication by a radial cutoff, into a low-frequency piece, a sum of dyadic annulus blocks, and a high-frequency remainder (`LogBound.lowP`, `midP`, `highP`, `telescope`). The annulus symbols $Q_{kl} = \psi \cdot m_{kl}$ are Schwartz functions. Dilation leaves the L^1 norm of the inverse-Fourier kernel unchanged (`integral_norm_fourierInv_dilate`), and a uniform kernel constant $A = \sum_{k,l} \int |\mathcal{F}^{-1}Q_{kl}|$ is formed (`LogBound.Akern`).

Low frequencies are estimated in terms of $\|f\|_2$, after the localization.

Dyadic blocks obey a scale-invariant bound: each block applied to a Schwartz function is controlled by A times a supremum of the function (or of $\operatorname{curl} f$, after the Helmholtz identity on the annulus, `LogBound.block_identity`). The number of dyadic scales that can be present before the H^3 energy is exhausted is comparable to $1 + \log(1 + \|f\|^2_{H^3})$. This is the source of the logarithm in (15).

High frequencies are estimated by the H^3 tail, which is absorbed into the same logarithmic factor and into an additive constant.

Assembly. Summing the nine component derivatives and taking the supremum over x_0 produces a constant C (an explicit combination of the kernel constant A, bump bounds, and the constants in `GradBound`) for which (15) holds. The function $X = 1 + \|f\|_2^2 + \|\omega\|_\infty\big(1 + \log(1 + \|f\|^2_{H^3})\big)$ is at least 1, so constant terms may be absorbed into a multiple of X. □

The inequality (15) is a pointwise estimate for a single smooth snapshot of finite H^3 energy. Combined with a Kato–Ponce bound on $\frac{d}{dt}\|u\|^2_{H^3}$ and a maximal-interval alternative in H^3, it yields the Beale–Kato–Majda continuation criterion: a uniform ceiling on $\|\omega\|_\infty$ prevents H^3 blow-up. That combination is formalized as `katoBKMMaximal_of_kato_of_log` (`R3BKMContinuation.lean:208`) and `r3BKMContinuation_of_kato_of_log` (:269). The log inequality is no longer a hypothesis. The remaining input is `KatoMaximalSolution`.

12 Kato infrastructure that is proved, and the remaining maximal-solution items

Definition 12.1 (Kato maximal solution)**.** The proposition `KatoMaximalSolution` ν u_0 (`Scratch/R3BKMContinuation.lean:76--84`) states that there exist a time set S, fields (u, p), and a constant $C_k \geq 0$, such that:

(i) for every T with $[0, T) \subseteq S$, the pair (u, p) is an `IsFeffermanSolutionOn` solution from u_0 on $[0, T)$;

(ii) $\|u(t)\|^2_{H^3}$ is finite for every $t \in S$, and the map $t \mapsto \|u(t)\|^2_{H^3}$ is continuous on S and right-differentiable, with
$$\frac{d}{dt}\|u\|^2_{H^3} \;\leq\; C_k \,\|\nabla u\|_\infty \,\|u\|^2_{H^3};$$

(iii) either $S = [0, \infty)$, or $S = [0, T^*)$ for some $T^* \in (0, \infty)$ and $\|u\|^2_{H^3}$ is unbounded on S.

The Lean docstring of this definition mentions uniqueness, persistence of H^s regularity with T^* independent of s, parabolic smoothing up to $t = 0$, energy equality, and a Kato–Ponce commutator estimate. The formal proposition is exactly the list (i)–(iii) above: one pair (u, p)

on a single interval S, Fefferman-class membership on every compactly contained subinterval, an H^3 energy inequality of Kato–Ponce type, and the H^3 blow-up alternative. Persistence for every s, uniqueness, and energy equality are not conjuncts of the proposition as stated.

The proposition is unsettled for general Fefferman data. It is proved at the zero datum: `katoMaximalSolution_zero` (`R3BKMContinuation.lean:254`).

12.1 Infrastructure that is kernel-checked

The following theorems are proved on the Fourier-side spaces $H^s :=$ `Hm s`, at commit `b497281f935f256d2a05372d5955296794f1d46d` on `feat/r3-bkm-continuation`, except where a contributing commit is named. Each depends only on `propext`, `Classical.choice` and `Quot.sound`.

Leray projector and heat semigroup. The Fourier-multiplier Leray projector is a contraction, a projection, and self-adjoint, and it kills gradients and reproduces divergence-free fields (`LerayHeat.norm_leray_le`, `leray_leray`, `inner_leray`, `leray_divFree`, `leray_eq_self`, `leray_grad`, `Scratch/LerayHeat.lean:186--220`). The heat semigroup $e^{t\Delta}$ is a contraction, is the identity for $t \le 0$, satisfies the semigroup law for $t, s \ge 0$, commutes with Leray, and is strongly continuous at $t = 0$ (`norm_heat_le`, `heat_of_nonpos`, `heat_heat`, `heat_leray`, `tendsto_heat_zero`, `continuous_heat`, `LerayHeat.lean:284--405`).

Equality of spatial and Fourier H^s energies. For a smooth field of finite spatial H^s energy, that energy equals the Fourier energy with weight `sobW s` (`SobolevFourier.sobolevEnergy_eq_fourierEnergy`, `Scratch/SobolevFourier.lean:219`). Fefferman decay implies that the spatial energy of every order is finite (`FeffermanSobolev.sobolevEnergy_lt_top_of_feffermanDecay`, `Scratch/FeffermanSobolev.lean:58`), and likewise the Fourier energy (:79).

Heat smoothing. The heat semigroup contracts Fourier H^s energy, and smooths $H^s \to H^{s+1}$ with a factor $1 + 1/(2\tau)$ for $\tau > 0$ (`HeatSobolev.fourierEnergy_heat_le`, `fourierEnergy_heat_succ_le`, `Scratch/HeatSobolev.lean:70,84`).

Duhamel integrals. The Duhamel integral of a continuous H^s-valued path obeys a norm bound and a Lipschitz estimate in the forcing, and is continuous in time (`Duhamel.norm_duh_le`, `duh_sub`, `norm_duh_sub_le`, `DuhamelUp.continuous_duh`). A smoothing Duhamel integral maps H^s into H^{s+1} with an explicit bound.

Tame product and the nonlinearity. For $s \ge 3$, Fourier-side convolution against a continuous bilinear form L satisfies

$$\|F \star [L]G\|_{H^s} \;\le\; C_s \|L\| \big(\|F\|_{H^s} \|G\|_{H^3} + \|F\|_{H^3} \|G\|_{H^s} \big)$$

(`ProductHm.norm_mulF_le`, `Scratch/ProductHm.lean:398`). The projected nonlinearity NSN : $H^s \to H^{s-1}$ obeys

$$\|\mathrm{NSN}(u)\|_{H^{s-1}} \;\le\; 18\, C_s\, \|u\|_{H^s}\, \|u\|_{H^3}$$

and a Lipschitz estimate in u (`NonlinearHm.norm_NSN_le`, `norm_NSN_sub_le`, `Scratch/NonlinearHm.lean:227,242`). Here NSN is defined at `NonlinearHm.lean:205`.

Local mild existence. For $\nu > 0$ and $u_0 \in H^3$ there is a time $T = \texttt{localTime}\,\nu\,u_0 > 0$ and a unique mild solution in the closed ball of radius $\texttt{rad}\,u_0$ in $C([0,T]; H^3)$, obtained as a Banach fixed point (`MildExistence.local_mild_existence`, `unique_mild_in_ball`, `Scratch/MildExistence.lean:470,454`).

Schwartz class and inverse Fourier regularity. A smooth Fefferman datum, complexified, is a Schwartz map (`FeffermanSchwartz.toSchwartz`, `Scratch/FeffermanSchwartz.lean:155`), and Fourier inversion holds pointwise (`fourierInv_fourier_cplx_of_fefferman`, :165). If a Fourier transform has finite H^s energy of every order, its inverse Fourier transform is C^∞ (`InverseSobolev.contDiff_fourierInv_of_all_energy`, `Scratch/InverseSobolev.lean:377`).

Initial energy. Fefferman decay implies $\int_{\mathbb{R}^3} |u_0|^2 < \infty$ (`R3BKM.lintegral_sq_lt_top_of_feffermanDecay`, `R3BKMContinuation.lean:98`).

Local persistence of H^s regularity. A mild solution in H^{k+1} remains a mild solution in that space on the H^3 local existence interval of its H^3 inclusion (`PersistenceHm.persist_on_H3_time`, `Scratch/PersistenceHm.lean:1842`; for Fefferman data, `FeffermanMild.persist_fefferman`, `Scratch/FeffermanMild.lean:112`). This is local persistence on the Kato existence time in H^3, for every $k \geq 2$. It is not persistence on a maximal interval T^* independent of s, which remains part of the open assembly in Section 12.2.

Time regularity of the mild path, and the mild equation. If $m \geq 2$ and $m + 2n \leq k$, the included mild path is C^n in time as an $\widehat{H}^m$-valued map on $[0, T)$ (`MildPathSmooth.contDiffOn_inclExt`, `Scratch/MildPathSmooth.lean:100`). Its time derivative equals $\nu\Delta u - N(u)$ in $\widehat{H}^m$ (`MildPathSmooth.mildSDeriv_eq_NSNsucc`, :44; equivalently `MildTimeDeriv.mildSDeriv_eq_lap`, `Scratch/MildTimeDeriv.lean:328`, for the definition `mildSDeriv` at :180).

Joint regularity of the realized mild field. The physical field `physicalMild` associated with a smooth Fefferman datum (`Scratch/FeffermanMild.lean:277`) is, on $\mathbb{R}^3 \times [0, T)$ with T the H^3 local time, jointly continuous (`MildJointSmooth.continuousOn_physicalMild`, `Scratch/MildJointSmooth.lean:54`), jointly C^1 (`contDiffOn_physicalMild_one`, :73), and jointly C^2 (`contDiffOn_physicalMild_two`, :108, commit `4722241ed9005aab3eb825a5a0adbbdb189e0a89`). The C^1 and C^2 statements include the one-sided time derivative at $t = 0$. Each snapshot is spatially C^∞ (`FeffermanMild.contDiff_physicalMild`, `FeffermanMild.lean:319`). Joint C^∞ regularity is not proved.

Kato–Ponce commutator and the conditional H^3 energy bound. The Gagliardo–Nirenberg inequality used by the H^3 commutator is proved (`KatoPonceCommutator.h3GagliardoNirenberg`, `Scratch/KatoPonceCommutator.lean:1725`, commit `a1a0e51f3f522384ff963c7b918439f6326c78ea`). The commutator estimate `H3CommutatorEstimate` itself is proved (`h3CommutatorEstimate`, :3320, commit `6f0a5ca9efd99a091d7b94f8ad5ae88f6111f0d6`; the statement of the proposition is at `KatoPonce.lean:516` and is unchanged). Consequently the H^3 energy bound holds with that estimate discharged: `katoPonce_h3_energy'` (:3426, commit `875b195b70645452ee2972ced302ddc5ead92db4`) has no `H3CommutatorEstimate` hypothesis. Partial-derivative commutation along a Fefferman-class solution is proved (`KatoPonceHypotheses.h3PartialCommutation`, `Scratch/KatoPonceHypotheses.lean:341`, commit `4fb2d6398b6418229669c40c461f4dbd99ac00ea`; from joint smoothness, `h3PartialCommutation_of_contDiff`, :333). The remaining hypotheses of `katoPonce_h3_energy'` are $0 \leq \nu$, membership in `IsFeffermanSolutionOn`, finite H^3 energy on $[0, T)$, `H3EnergyRightDerivativeFormula` (`KatoPonce.lean:439`), and `H3IBPVanishingFlux` (:501). The last two are open.

Leray pressure. A Leray pressure is constructed from an H^s Fourier representative for $s \geq 3$ (`LerayPressure.lerayPressure`, `Scratch/LerayPressure.lean:306`, with the Fourier-side field

`lerayPressureHm` at :202), and is C^n when $n+2 \le s$ (`contDiff_lerayPressure`, :347, commit `f584cfa7c0fc5bd6c0433e5766a58efa8df454ab`). This does not by itself identify the pressure in the pointwise momentum equation, nor prove that the mild field is divergence-free.

Whole-space integral of a divergence. If F is C^1 on $\mathbb{R}^3$ and both F and $\operatorname{div} F$ are integrable, then $\int_{\mathbb{R}^3} \operatorname{div} F = 0$ (`DivergenceWholeSpace.integral_div_eq_zero`, `Scratch/DivergenceWholeSpace.lean:302`, commit `b497281f935f256d2a05372d5955296794f1d46d`; the compactly supported case is `integral_div_eq_zero_of_hasCompactSupport`, :182). This is the whole-space flux identity under those integrability hypotheses. It does not discharge `H3IBPVanishingFlux`, whose remaining obstruction is the integrability of the relevant products and derivatives (a Plancherel identification of physical L^2 norms of derivatives of the inverse-Fourier representative is not proved).

These theorems are the infrastructure. They do not, separately or together, prove `KatoMaximalSolution` for general data.

12.2 Remaining items

The following pieces of Definition 12.1 are unsettled for general Fefferman data, at commit `b497281f935f256d2a05372d5955296794f1d46d`.

(i) **Joint C^∞.** That the realized mild field is jointly C^∞ on $\mathbb{R}^3 \times [0,T)$. Joint continuity, joint C^1 and joint C^2, including the one-sided time derivative at $t=0$, are proved. The step from C^2 to every finite order is not complete.

(ii) **Incompressible momentum equation.** That the realized mild field is divergence-free and satisfies the pointwise Navier–Stokes momentum equation with a pressure (and a one-sided time derivative at $t=0$). A Leray pressure is defined and proved smooth; the identification of that pressure with the momentum equation, and $\operatorname{div} u = 0$ along the mild path, are not proved.

(iii) **Energy inequality.** The comparison $\int |u(t)|^2 \le \int |u_0|^2$ in the Fefferman class, as a lower Lebesgue integral, for every snapshot of the realized mild field.

(iv) **Energy derivative and vanishing flux.** The remaining hypotheses of `katoPonce_h3_energy'` (and therefore of Definition 12.1(ii) along a Fefferman-class solution): `H3EnergyRightDerivativeFormula` and `H3IBPVanishingFlux`. The latter is not implied by `IsFeffermanSolutionOn`: that class fixes the pressure only up to a function of time and puts no decay on later snapshots, so the required integrability and flux identities are not controlled.

(v) **Maximal interval.** Assembly of a maximal S with the H^3 blow-up alternative of Definition 12.1(iii). Persistence of H^s regularity on that maximal interval, with T^* independent of s, is likewise not proved.

A classical reference for local existence in H^s and the blow-up alternative is Kato [3]. That citation is unformalized background; it is not a Lean theorem and is not used as a hidden hypothesis of Theorems 4.3 or 4.4. Those theorems take `KatoMaximalSolution` as an explicit, still-open input.

Snapshot decay propagation (Section 7) is not a conjunct of `KatoMaximalSolution` as stated. Given the local persistence and inverse-Fourier regularity already proved, the remaining analytic step for later snapshots is vanishing of vorticity at infinity. That step is listed as a separate open item in Section 13.

13 Open hypotheses

Every item below is marked proved, disproved, or unsettled. No item is described as probable.

Item	Status	Lean pointer; rate files at `5d8e6d6`, Kato files at `b497281`
`Clay_A_Fefferman`	unsettled	official statement, `RepairedTetherRate.lean:46`
`FrohmanianTetherRateLipschitz` for general Fefferman data, $\nu > 0$	unsettled	`TetherRateLipschitz.lean:44`
same, at $u_0 = 0$	proved	`frohmanianTetherRateLipschitz_zero`, :263
`FrohmanianTetherRateSmooth` for general data	unsettled	`RepairedTetherRate.lean:74`
same, at $u_0 = 0$	proved	`frohmanianTetherRateSmooth_zero`, :170
`KatoMaximalSolution` for general data	unsettled	`R3BKMContinuation.lean:76`; remaining items in §12.2
same, at $u_0 = 0$	proved	`katoMaximalSolution_zero`, :254
`BKMLogInequality`	proved	`bkmLogInequality`, `LogBoundLocal.lean:659`
Lipschitz rate + Kato $\Rightarrow$ Clay (A)	proved (implication)	`clay_A_fefferman_of_rateLipschitz_of_kato`, `TetherRateLipschitz.lean:251`
Smooth rate + Kato $\Rightarrow$ Clay (A)	proved (implication)	`clay_A_fefferman_of_rate_of_kato`, `LogBoundLocal.lean:719`
Smooth $\Rightarrow$ Lipschitz, with a locally bounded right derivative	proved	`frohmanianTetherRateLipschitz_of_smooth_of_locallyBddRightDeriv`, :361
Smooth $\Rightarrow$ Lipschitz, with no extra bound	unsettled	§4.3
Comparison ceiling, Lipschitz / a.e. form	proved	`comparison_ode_stability_lipschitz`, :82
Bochner, transport cancellation, viscous sign at a maximum	proved	`TetherRateDerivation.lean:78,147,256`
Cubic assembly as an implication	proved	`cubic_rate_of_strain_and_hsSq`, :390
Parabolic scaling of M and M'	proved	`Mω_rescaleFlow`, :513; `hasDerivAt_Mω_rescale_of_hasDerivAt`, :548
Universal $\|\nabla u\| \le C\|\omega\|_\infty$ for arbitrary fields	disproved	`not_forall_strain_le_sup_vorticity`
`StrainControlledByVorticity` along Fefferman solutions (inequality 3)	unsettled	`TetherRateDerivation.lean:465`
`DirectionCoercivityAtMax` along solutions (inequality 4)	unsettled	:474
Static coercivity for arbitrary snapshots	not formalized in the cited commits	§10
Attainment of $\|\omega\|$ given vanish-at-infinity, or given snapshot Fefferman decay	proved	`VorticityMaxAttained.lean:321,335`
Attainment at $t = 0$ for Fefferman data	proved	:350
`IsFeffermanSolutionOn` $\Rightarrow$ attainment, as written	disproved	`not_isFeffermanSolutionOn_implies_vorticityMaximumAttained`, :569
Snapshot decay propagation for $t > 0$	unsettled	§7
Lipschitz / Rademacher / Danskin for M, under a uniform bound on $\partial_t\omega$ and attainment	proved	`VorticityMaxAttained.lean:631,644,671,751`

Item	Status	Lean pointer; rate files at `5d8e6d6`, Kato files at `b497281` (continued)
That uniform bound on $\partial_t\omega$, for general Fefferman solutions	unsettled	`VorticityTimeDerivBddOn`, :586
`DifferentiableMaximizer` (a C^1 maximizing path)	unsettled, and not used by the Lipschitz rate	`TetherRateDerivation.lean:454`
`FrohmanianTetherRate` as originally written	disproved	`not_forall_frohmanianTetherRate`
`h3GagliardoNirenberg`	proved	`KatoPonceCommutator.lean:1725`, commit `a1a0e51`
`h3CommutatorEstimate`	proved	`KatoPonceCommutator.lean:3320`, commit `6f0a5ca`; statement at `KatoPonce.lean:516`
`katoPonce_h3_energy'` (commutator discharged)	proved, with remaining hypotheses	`KatoPonceCommutator.lean:3426`, commit `875b195`
`h3PartialCommutation`	proved	`KatoPonceHypotheses.lean:341`, commit `4fb2d63`
`H3EnergyRightDerivativeFormula`	unsettled	`KatoPonce.lean:439`
`H3IBPVanishingFlux`	unsettled; not implied by `IsFeffermanSolutionOn`	`KatoPonce.lean:501`
Local H^s persistence on the H^3 existence interval	proved	`persist_on_H3_time`, `PersistenceHm.lean:1842`; `persist_fefferman`, `FeffermanMild.lean:112`
C^n in time of the mild path in $\widehat{H}^m$	proved	`contDiffOn_inclExt`, `MildPathSmooth.lean:100`
$\partial_t u = \nu\Delta u - N(u)$ in $\widehat{H}^m$	proved	`mildSDeriv_eq_NSNsucc`, `MildPathSmooth.lean:44`
Joint C^0, C^1, C^2 of `physicalMild`, including the one-sided derivative at $t = 0$	proved	`MildJointSmooth.lean:54,73,108`, commit `4722241`
Joint C^∞ of `physicalMild`	unsettled	§12.2
`lerayPressure` and `contDiff_lerayPressure`	proved	`LerayPressure.lean:306,347`, commit `f584cfa`
$\int_{\mathbb{R}^3} \operatorname{div} F = 0$ for C^1 integrable F with integrable divergence	proved	`integral_div_eq_zero`, `DivergenceWholeSpace.lean:302`, commit `b497281`
Divergence-free and pointwise momentum equation for the mild field	unsettled	§12.2
Energy inequality for the realized mild field	unsettled	§12.2
Maximal interval and H^3 blow-up alternative	unsettled	§12.2
Kato infrastructure of §12.1	proved	cited there, at `b497281`
T1–T10 of the gaps ledger	unsettled	§14; none is discharged here

14 Gaps T1–T10, restated from the strain-kernel record

This section restates the named gaps T1–T10 from the earlier working notes [7], themselves an update of `10.5281/zenodo.22835069`, and inserts them as a hypotheses-and-gaps ledger. It does not derive any estimate that would close a named gap. Those notes concerned a strain-kernel identification $A(\theta, \omega) = (\theta \times \omega) \otimes \theta + \theta \otimes (\theta \times \omega)$ and a prefactor $\kappa = C_{\mathrm{CZ}}(3) = 3/2$ on the torus $\mathbb{T}^3$. The present paper's open inequalities 3 and 4 are posed on $\mathbb{R}^3$ along Fefferman solutions;

they are related in subject to T1 and to the missing cubic sink, but they are not the same propositions, and no identification of them with T1–T10 is claimed.

The body of the published PDF of the source record never prints the tokens T1, …, T10. The obstruction section is written as Lemmas 4.1–4.8. The abstract and deposit description say that named gaps T1–T10 (Lemmas 4.1–4.6) record why a bound on $\|S\|_\infty$ is not derived there, and they add two scope statements that are not numbered as lemmas. The following is a ledger reading of that abstract phrase, not a typographic fact of the PDF.

Label	Source in 22835069	What it blocks
T1	Lemma 4.1, strain endpoint	a solution-independent space X and constant C_* with $\|S\|_{L^\infty} \le C_*\|\omega\|_X$
T2	Lemma 4.2, stretching identification	$\int\langle\omega_\varepsilon, (\omega_\varepsilon\cdot\nabla)u_\varepsilon\rangle\phi_\varepsilon \le C_* M_\varepsilon I_4(\phi_\varepsilon) + C_\varepsilon$ with commutators compatible with a data-only majorant
T3	Lemma 4.3, transported weight	a weight ϕ_ε whose L^∞ ODE coefficients depend only on $\|u_0\|_{H^s}$ and ν (if $\phi \equiv 1$, the tether is ornamental and the problem moves onto Lemma 4.2)
T4	Lemma 4.4, absorption constant	C_{abs} depending only on $\kappa = 3/2$, C_{Sob}, $\lambda(\mathbb{T}^3)$, and ν
T5	Lemma 4.5, non-circular majorant	a comparison function Y whose ODE coefficients are independent of $\{M(t) : t \ge 0\}$, with $M \le Y$
T6	Lemma 4.6, mollifier limit	commutator constants that remain controlled as $T \uparrow T_*$, depending only on u_0 and ν
T7	Lemma 4.7, Jacobi / cocycle of B	verification of a tethered bracket that is simultaneously degenerate on H, Jacobi, and a source of a new negative term in $\frac{d}{dt}M$ (in tension; vacated if B is dropped)
T8	Lemma 4.8, degeneracy versus feedback	the dichotomy: a Poisson correction with $B(\cdot, H) = 0$ cannot supply a new term in the vorticity equation; a nonzero correction changes the PDE
T9	abstract / description	an L^∞ bound on S on the whole torus is emptiness of the singular set, not $\mathbb{P}^1$-nullity (Caffarelli–Kohn–Nirenberg [5])
T10	abstract / description	the two-point strain pairing that would discharge Constantin–Fefferman [6] is not free

T1–T6 are the six hypotheses of Theorem 2.1 of that record. Section 6 of the source says that those analytic pieces are the same missing a priori bound under different names. Proving them would prove that theorem unconditionally, which is a Clay-type statement on $\mathbb{T}^3$. No rearrangement of the names discharges the estimate. In the present work, T1–T10 are all **unsettled**.

T1 (Lemma 4.1, strain endpoint). There exist a Banach space $X(\mathbb{T}^3)$ and a constant C_* such that $\|S\|_{L^\infty} \le C_*\|\omega\|_X$ for every smooth divergence-free ω, with S the deformation tensor of $u = K_{\mathrm{BS}} * \omega$. The Calderón–Zygmund operator $\omega \mapsto S$ is bounded on $L^p(\mathbb{T}^3)$ for $1 < p < \infty$ and is not bounded on L^∞. The estimate $\|S\|_{L^\infty} \le \frac{3}{2}\|\omega\|_{L^\infty}$ is therefore false as a general statement. A Kozono–Taniuchi or $\dot{B}^0_{\infty,1}$ substitute changes the bookkeeping. Until the endpoint is fixed, Lemma 4.2 cannot feed a Young absorption. Status: unsettled. Related to, but not identified with, `StrainControlledByVorticity` on $\mathbb{R}^3$.

T2 (Lemma 4.2, stretching identification). On a classical solution, after mollification, the displayed integral bound of the table holds, with $C_\varepsilon \to 0$ in the sense used to pass to the limit, and with C_* the constant of T1. The algebraic identity $\langle\omega, (\omega\cdot\nabla)u\rangle = \langle\omega, S\omega\rangle$ is filled. The missing piece is the same L^∞ bound, plus mollifier commutators whose constants may depend on $\|u\|_{C^2}$ and therefore cannot be fed into a majorant whose coefficients may depend only on u_0 and ν. Status: unsettled.

T3 (Lemma 4.3, transported weight). There exists a family of nonnegative scalars ϕ_ε such that $\|\phi_\varepsilon(t)\|_{L^\infty}$ is dominated by a function of t whose ODE coefficients depend only on $\|u_0\|_{H^s}$ and ν. Linear transport with $\operatorname{div} u = 0$ conserves $\|\phi\|_{L^\infty}$ and is filled. If the weight grows in response to stretching, the ODE inherits the unknown C_*. Taking $\phi \equiv 1$ makes the tether ornamental and moves the problem onto T2. Status: unsettled.

T4 (Lemma 4.4, absorption constant). The constant C_{abs} in the schematic inequality for $\frac{d}{dt}S_\varepsilon$ depends only on $\kappa = 3/2$, C_{Sob}, $\lambda(\mathbb{T}^3)$, and ν. Until T1 supplies a solution-independent C_*, C_{abs} is not a function of u_0 and ν alone. Status: unsettled.

T5 (Lemma 4.5, non-circular majorant). There exists a scalar $Y \in C^1([0,\infty);[0,\infty))$ whose ODE $Y' = F(Y)$ has coefficients independent of $\{M(t) : t \geq 0\}$, such that $M(t) \leq Y(t)$ on the existence interval. Scalar ODE comparison, including a clipped cubic $F(y) = C_{\mathrm{abs}}y^2 - \kappa'' y^3$ with a global barrier, is filled as an implication; the Lean comparison lemmas of Section 5 are of this type. The missing piece is the differential inequality $M' \leq F(M)$ with an F built only from u_0 and ν. Inserting $\sup M$ into C_{abs} is circular; modifying the PDE is inadmissible. Status: unsettled, and the same core as the open rate.

T6 (Lemma 4.6, mollifier limit). $S_\varepsilon(t) \to S(t)$ and the commutators vanish as $\varepsilon \to 0$, with constants compatible with T5. Passage on a compact classical interval $[0,T] \subset [0,T_*)$ is filled. Using those bounds to continue past a putative singularity requires constants that remain controlled as $T \uparrow T_*$. Status: unsettled.

T7 (Lemma 4.7, Jacobi / cocycle) and T8 (Lemma 4.8, degeneracy versus feedback). If a bilinear correction B is part of the argument, then $d^2B = 0$ on the reduced coadjoint orbit, and a correction used as dynamics satisfies $B(F,H) = 0$. If $B(\cdot,H) = 0$, the Hamiltonian evolution is exactly Euler and no extra term appears in the vorticity equation. If $B(\cdot,H) \neq 0$, the evolution is not Navier–Stokes. If B is dropped and ϕ is an ordinary weight, T7 is vacated. Status: unsettled as geometry; filled as logic in the source record. The reductions of Section 4 do not use a Poisson correction B.

T9 and T10 (abstract scope statements). An L^∞ bound on S on the whole torus is emptiness of the singular set, not $\mathbb{P}^1$-nullity in the sense of Caffarelli–Kohn–Nirenberg. The two-point strain pairing that would discharge Constantin–Fefferman is not free. Neither sentence is a proof of a bound, and neither is weakened here. Status: unsettled as claims that would close a regularity argument; recorded as scope.

15 Earlier tether formulations: refutation and vacuity

The original proposition `FrohmanianTetherRate` quantified over a solution class `IsPhysicalNSSolution` that does not require spatial differentiability. Mathlib's Fréchet derivative is 0 at points of non-differentiability. A steady, finite-energy, continuous field with a nowhere-differentiable Takagi component [4] therefore formally satisfies the equations for every viscosity, while its vorticity is unbounded. Consequently

$$\neg\, \forall u_0\, \nu, \quad \texttt{FrohmanianTetherRate}\ u_0\ \nu,$$

and the companion bound `FrohmanianTetherBound` fails for the same field at every ν. These are `TetherRateCounterexample.not_forall_frohmanianTetherRate` and `not_frohmanianTetherBound_U` (`Scratch/TetherRateCounterexample.lean:605,610`). Status: kernel-checked disproved, as written.

On continuous $\mathbb{Z}^3$-periodic data the original rate holds, and for nonzero data it holds only vacuously (`frohmanianTetherRate_of_periodic`, `Scratch/TetherRateStatus.lean`). The repaired rates of Definitions 4.1 and 4.2 quantify only over Fefferman's smooth class, which excludes the Takagi counterexample.

In the repository `NS_Millennium_Proof`, the binder `hDI` of the earlier conditional theorem `frohmanian_tether_theorem` cannot be inhabited; that theorem is vacuously true. This is recorded in the earlier reduction note and is not re-proved here.

The repaired rates are not refuted by the Takagi example, and they are not proved for general Fefferman data.

16 Reproducibility

Author and toolchain. Author: Benjamin Stanley Frohman (B.S. Frohman), ORCID `0009-0006-7068-3718`, `frohmanbenjamin@gmail.com`, Independent Researcher, Texas, United States. Lean `v4.30.0-rc1` (binary report: `Lean (version 4.30.0-rc1, x86_64-unknown-linux-gnu, commit 714601baf118066cbf3f282361339c6d06665b2a, Release)`). Mathlib revision `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7` (the lake manifest pin `v4.30.0-rc1`). Allowed axioms: `propext`, `Classical.choice`, `Quot.sound` only.

Commits. Repository `github.com/BenFrohman/frohmanian-tether-sdiff`. The two branches below diverged at `5dd8458aa85cd118962b8b7ef2f37480cf2cd0f7`.

- Branch `feat/r3-bkm-continuation`, commit `b497281f935f256d2a05372d5955296794f1d46d`: the Kato infrastructure of Section 12.1 and the pin for this document's Lean code. This is the commit against which line numbers for Kato files were checked.
- Branch `feat/tether-rate-derivation`, commit `5d8e6d679666d4cbd785326422695165655ab1ef`: rate hypotheses, comparison lemma, attainment, scaling, BKM inequality, and both Clay (A) reductions. The Lipschitz rate was introduced at `c9ffdeb1bfe8b7252bf93e6bd97f5455ed5afff7`; the tip `5d8e6d6` updates docstrings only.

Neither branch is a publication, and this document is not a Zenodo deposit.

One rebuild command. From a checkout of `feat/r3-bkm-continuation` at `b497281`, with the Mathlib package already present at revision `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7` (do not rebuild Mathlib),

```
lake env lean scripts/PrintScratchAxioms.lean
```

That command is the Scratch axiom audit run by the repository's Lean certificate workflow. It does not rebuild Mathlib.

Axiom audit of the printed theorems. The Scratch axiom audit at the pinned Kato commit, from GitHub Actions run `43158531834` on commit `b497281f935f256d2a05372d5955296794f1d46d`, reports `SCRATCH AUDIT SUMMARY: theorems audited: 102; clean: 102; violations: 0` and `SCRATCH AUDIT PASS: only propext, Classical.choice, Quot.sound; no sorryAx`. The listing includes the joint C^2 headlines and the whole-space divergence theorems. The verbatim output is Appendix A.

A companion listing of 69 rate, comparison, attainment and reduction theorems, each depending only on `propext`, `Classical.choice` and `Quot.sound`, was produced at `c9ffdeb1bfe8b7252bf93e6bd97f5455ed5afff7` on `feat/tether-rate-derivation` (Appendix B). The tip `5d8e6d6` changes docstrings only.

References

[1] C. L. Fefferman, *Existence and smoothness of the Navier–Stokes equation*, in: The Millennium Prize Problems, Clay Mathematics Institute, 2006, pp. 57–67.

[2] J. T. Beale, T. Kato, A. Majda, Remarks on the breakdown of smooth solutions for the 3-D Euler equations, *Comm. Math. Phys.* 94 (1984), 61–66.

[3] T. Kato, Strong L^p-solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions, *Math. Z.* 187 (1984), 471–480.

[4] T. Takagi, A simple example of the continuous function without derivative, *Proc. Phys.-Math. Soc. Japan* 1 (1903), 176–177.

[5] L. Caffarelli, R. Kohn, L. Nirenberg, Partial regularity of suitable weak solutions of the Navier–Stokes equations, *Comm. Pure Appl. Math.* 35 (1982), 771–831.

[6] P. Constantin, C. Fefferman, Direction of vorticity and the problem of global regularity for the Navier–Stokes equations, *Indiana Univ. Math. J.* 42 (1993), 775–789.

[7] B.S. Frohman, Identified strain kernel $A(\theta, \omega)$ and a conditional regularity criterion for 3D Navier–Stokes ($\kappa = 3/2$): named open hypotheses, gaps T1–T10, and the current Lean implications, Zenodo (2026), doi:10.5281/zenodo.22835069, working-notes update as drafted 11 November 2026.

Appendix: axiom output

A. Scratch axiom audit at `b497281`

The following is the verbatim output of `lake env lean scripts/PrintScratchAxioms.lean` from GitHub Actions run `43158531834`, on `feat/r3-bkm-continuation` at commit `b497281f935f256d2a05372d5955296794f1d46d`, with Mathlib already present at revision `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`.

```
'LogBoundLocal.bkmLogInequality' depends on axioms: [propext, Classical.choice, Quot.sound]
'LogBoundLocal.clay_A_fefferman_of_rate_of_kato' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildExistence.local_mild_existence' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProductHm.norm_mulF_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProductHm.eLpNorm_conv_sq_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProductHm.weighted_conv_tame' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProductHm.weighted_conv_le_pow' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearHm.norm_NSN_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearHm.norm_NSN_sub_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'InverseSobolev.contDiff_fourierInv_of_all_energy' depends on axioms: [propext, Classical.choice, Quot.sound]
'InverseSobolev.integral_norm_coeFn_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanSchwartz.fourierInv_fourier_cplx_of_fefferman' depends on axioms: [propext, Classical.choice, Quot.sound]
'PersistenceHm.persist_on_minTime' depends on axioms: [propext, Classical.choice, Quot.sound]
'PersistenceHm.apriori_double' depends on axioms: [propext, Classical.choice, Quot.sound]
'PersistenceHm.persist_chunk_bound' depends on axioms: [propext, Classical.choice, Quot.sound]
'PersistenceHm.persist_on_H3_time' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonce.katoPonce_h3_energy' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanMild.physicalOfHm_toHmFeff' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanMild.persist_fefferman' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanMild.contDiff_physicalOfHm' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanMild.physicalOfHm_incl' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanMild.inclMap_persistPath_unique' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanMild.physicalMild_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanMild.contDiff_physicalMild' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceCommutator.convectiveCommutator_order_one' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceCommutator.abs_integral_commutator_order_one' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceCommutator.integral_gradNormSq_sq' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceCommutator.integral_gradNormSq_sq_le_of_hasCompactSupport' depends on axioms: [propext, Classical.choice, Quot
'KatoPonceCommutator.h3GagliardoNirenberg' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceCommutator.integral_gradNormSq_sq_le_of_energy' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceCommutator.abs_integral_commutator_order_two' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceCommutator.abs_integral_commutator_order_three' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceCommutator.h3CommutatorEstimate' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceCommutator.katoPonce_h3_energy'' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceHypotheses.wordDeriv_laplacian_eq' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceHypotheses.wordDeriv_pressureGradient_eq' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceHypotheses.div_wordDeriv_eq' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceHypotheses.h3PartialCommutation_of_contDiff' depends on axioms: [propext, Classical.choice, Quot.sound]
'KatoPonceHypotheses.h3PartialCommutation' depends on axioms: [propext, Classical.choice, Quot.sound]
'HeatDeriv.lapH' depends on axioms: [propext, Classical.choice, Quot.sound]
'HeatDeriv.hasDerivWithinAt_gauss' depends on axioms: [propext, Classical.choice, Quot.sound]
'HeatDeriv.hasDerivWithinAt_heatH' depends on axioms: [propext, Classical.choice, Quot.sound]
'HeatDeriv.hasDerivWithinAt_heatH_visc' depends on axioms: [propext, Classical.choice, Quot.sound]
'IBPWholeSpace.integral_partial_mul' depends on axioms: [propext, Classical.choice, Quot.sound]
'IBPWholeSpace.integral_partial_inner' depends on axioms: [propext, Classical.choice, Quot.sound]
'DuhamelDeriv.duh_cocycle' depends on axioms: [propext, Classical.choice, Quot.sound]
'DuhamelDeriv.tendsto_duhνSame_avg' depends on axioms: [propext, Classical.choice, Quot.sound]
'DuhamelDeriv.hasDerivWithinAt_duhνSame_Ici' depends on axioms: [propext, Classical.choice, Quot.sound]
'DuhamelDeriv.incl_duhν_to_duhνSame' depends on axioms: [propext, Classical.choice, Quot.sound]
'DuhamelDeriv.incl_heatH_of_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'DuhamelDeriv.incl_duhνSame_of_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildTimeDeriv.incl_duhHigh_to_duhνSame' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildTimeDeriv.mildS_eq_duhνSame' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildTimeDeriv.hasDerivWithinAt_mildS_incl' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildTimeDeriv.continuous_mildSDeriv' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildTimeDeriv.hasDerivAt_mildS_incl' depends on axioms: [propext, Classical.choice, Quot.sound]
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'MildTimeDeriv.hasDerivWithinAt_mildS_incl_Ico' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildTimeDeriv.contDiffOn_one_inclExt' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildTimeDeriv.lapH_heatH' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildTimeDeriv.mildSDeriv_eq_lap' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearSmooth.isBoundedBilinear_mulF' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearSmooth.contDiff_mulF' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearSmooth.contDiff_mulF_quad' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearSmooth.contDiff_divTensor' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearSmooth.contDiff_NSN' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearSmooth.contDiff_NSNsucc' depends on axioms: [propext, Classical.choice, Quot.sound]
'InclNSN.incl_mulF_of_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'InclNSN.incl_dH_of_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'InclNSN.incl_NSN_of_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'InclNSN.coeFn_NSNsucc' depends on axioms: [propext, Classical.choice, Quot.sound]
'InclNSN.incl_NSNsucc' depends on axioms: [propext, Classical.choice, Quot.sound]
'InclNSN.incl_incl_NSNsucc' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildPathSmooth.incl_nsnExt_eq_NSNsucc' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildPathSmooth.mildSDeriv_eq_NSNsucc' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildPathSmooth.derivWithin_inclExt_eq_NSNsucc' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildPathSmooth.contDiff_lapH' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildPathSmooth.contDiffOn_inclExt' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildPathSmooth.contDiffOn_inclExt_of_le_two' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvBCF.isBoundedLinearMap_fourierInvBCF' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvBCF.continuous_uncurry_fourierInv' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvBCF.continuousOn_uncurry_clm_fourierInvBCF' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvBCF.contDiffOn_fourierInvBCF' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildJointSmooth.continuousOn_physicalMild' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.hasFDerivAt_fourierInvBCF' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.fourierInvBCF_eq_integral_spatFDeriv' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.hasFDerivAt_uncurryFourierInvBCF' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.memLp_coe_Hm' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.memLp_lerayPressureHm' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.abs_rieszCoeff_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.rieszSym_norm_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.norm_rieszMul_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.norm_lerayPressureHm_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.continuous_lerayPressure' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.abs_lerayPressure_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.bddAbove_lerayPressure' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.contDiff_physicalInv' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.contDiff_lerayPressure' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.contDiff_two_lerayPressure' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayPressure.contDiff_five_lerayPressure' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.continuous_jointFDeriv' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.contDiff_one_uncurryFourierInvBCF' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.contDiffOn_uncurryFourierInvBCF_path' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.contDiffOn_uncurry_clm_fourierInvBCF_one' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildJointSmooth.contDiffOn_physicalMild_one' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.hasFDerivAt_evMap' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.continuous_evFDeriv' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.contDiff_one_evMap' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.contDiff_two_uncurryFourierInvBCF' depends on axioms: [propext, Classical.choice, Quot.sound]
'FourierInvFDeriv.contDiffOn_uncurryFourierInvBCF_path_two' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildJointSmooth.contDiffOn_physicalMild_two' depends on axioms: [propext, Classical.choice, Quot.sound]
'DivergenceWholeSpace.integral_div_eq_zero_of_hasCompactSupport' depends on axioms: [propext, Classical.choice, Quot.sou
'DivergenceWholeSpace.integral_div_eq_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
SCRATCH AUDIT SUMMARY: theorems audited: 102; clean: 102; violations: 0
SCRATCH AUDIT PASS: only propext, Classical.choice, Quot.sound; no sorryAx
```

B. Rate, comparison, attainment and reduction theorems at `c9ffdeb`

The following is the verbatim output of a companion `#print axioms` listing run on `feat/tether-rate-derivation` at `c9ffdeb1bfe8b7252bf93e6bd97f5455ed5afff7`, with Mathlib already present at revision `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`.

```
'RepairedTether.comparison_ode_stability_lipschitz' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.ceiling_of_rateLipschitz' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.clay_A_fefferman_of_frohmanianTetherRateLipschitz' depends on axioms: [propext, Classical.choice, Quot.s
'RepairedTether.clay_A_fefferman_of_rateLipschitz_of_kato' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.frohmanianTetherRateLipschitz_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.frohmanianTetherRateLipschitz_of_smooth_of_locallyBddRightDeriv' depends on axioms: [propext, Classical.
'RepairedTether.lipschitzOnWith_of_bdd_right_deriv' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.majorant_vector_field_antitoneOn' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.clay_A_fefferman_of_frohmanianTetherRateSmooth' depends on axioms: [propext, Classical.choice, Quot.soun
'RepairedTether.ceiling_of_rate' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.frohmanianTetherRateSmooth_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.r3BKMContinuationAt_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.feffermanDecay_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.clay_A_fefferman_instance_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'RepairedTether.frohmanianTetherRateSmoothGlobal_of_no_global_solution' depends on axioms: [propext, Classical.choice, Q
'TetheredLyapunov.comparison_ode_stability_Icc_right' depends on axioms: [propext, Classical.choice, Quot.sound]
'LogBoundLocal.bkmLogInequality' depends on axioms: [propext, Classical.choice, Quot.sound]
'LogBoundLocal.clay_A_fefferman_of_rate_of_kato' depends on axioms: [propext, Classical.choice, Quot.sound]
'R3BKM.clay_A_fefferman_of_rate_of_kato_of_log' depends on axioms: [propext, Classical.choice, Quot.sound]
'R3BKM.katoMaximalSolution_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'R3BKM.katoBKMMaximal_of_kato_of_log' depends on axioms: [propext, Classical.choice, Quot.sound]
'R3BKM.r3BKMContinuation_of_kato_of_log' depends on axioms: [propext, Classical.choice, Quot.sound]
'R3BKM.lintegral_sq_lt_top_of_feffermanDecay' depends on axioms: [propext, Classical.choice, Quot.sound]
'R3BKM.log_gronwall' depends on axioms: [propext, Classical.choice, Quot.sound]
'VorticityMaxAttained.exists_isMaxOn_of_tendsto_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'VorticityMaxAttained.vorticity_tendsto_zero_of_feffermanDecay' depends on axioms: [propext, Classical.choice, Quot.soun
'VorticityMaxAttained.exists_vorticityMaximumAttained_of_tendsto_zero' depends on axioms: [propext, Classical.choice, Qu
'VorticityMaxAttained.exists_vorticityMaximumAttained_of_snapshot_feffermanDecay' depends on axioms: [propext, Classical
'VorticityMaxAttained.exists_isMaxOn_vorticity_at_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'VorticityMaxAttained.not_isFeffermanSolutionOn_implies_vorticityMaximumAttained' depends on axioms: [propext, Classical
'VorticityMaxAttained.lipschitzOnWith_Mω_of_timeDerivBdd' depends on axioms: [propext, Classical.choice, Quot.sound]
'VorticityMaxAttained.ae_differentiableAt_Mω_of_timeDerivBdd' depends on axioms: [propext, Classical.choice, Quot.sound]
'VorticityMaxAttained.danskin_eq_of_differentiable' depends on axioms: [propext, Classical.choice, Quot.sound]
'VorticityMaxAttained.danskinRateAt_of_differentiable' depends on axioms: [propext, Classical.choice, Quot.sound]
'VorticityMaxAttained.ae_danskinRateAt_of_timeDerivBdd' depends on axioms: [propext, Classical.choice, Quot.sound]
'VorticityMaxAttained.isFeffermanSolutionOn_arctanShear' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.bochner_identity' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.transport_vanishes_at_local_max' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.viscPair_le_neg_hsSq_at_local_max' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.scalarLaplacian_nonpos_of_isLocalMax' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.envelope_along_differentiable_maximizer' depends on axioms: [propext, Classical.choice, Quot.sound
'TetherRateDerivation.danskin_difference_quotient' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.stretching_le_strain_mul_norm_sq' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.cubic_rate_of_stretching_and_visc' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.cubic_rate_of_strain_and_hsSq' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.fefferman_two_sided_eq_interior' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.vorticity_rescaleFlow' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.Mω_rescaleFlow' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.hasDerivAt_Mω_rescale_of_hasDerivAt' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.rate_form_rescale_iff' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateDerivation.hasDerivAt_norm_of_ne_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
'StrainBoundRefutation.not_forall_strain_le_sup_vorticity' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateCounterexample.not_forall_frohmanianTetherRate' depends on axioms: [propext, Classical.choice, Quot.sound]
'TetherRateCounterexample.not_frohmanianTetherBound_U' depends on axioms: [propext, Classical.choice, Quot.sound]
'SobolevFourier.sobolevEnergy_eq_fourierEnergy' depends on axioms: [propext, Classical.choice, Quot.sound]
'HeatSobolev.fourierEnergy_heat_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'HeatSobolev.fourierEnergy_heat_succ_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanSobolev.sobolevEnergy_lt_top_of_feffermanDecay' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanSchwartz.fourierInv_fourier_cplx_of_fefferman' depends on axioms: [propext, Classical.choice, Quot.sound]
'InverseSobolev.contDiff_fourierInv_of_all_energy' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProductHm.norm_mulF_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearHm.norm_NSN_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'NonlinearHm.norm_NSN_sub_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildExistence.local_mild_existence' depends on axioms: [propext, Classical.choice, Quot.sound]
'MildExistence.unique_mild_in_ball' depends on axioms: [propext, Classical.choice, Quot.sound]
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'GradBound.bddAbove_norm_fderiv' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayHeat.norm_leray_le' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayHeat.heat_heat' depends on axioms: [propext, Classical.choice, Quot.sound]
'LerayHeat.tendsto_heat_zero' depends on axioms: [propext, Classical.choice, Quot.sound]
```